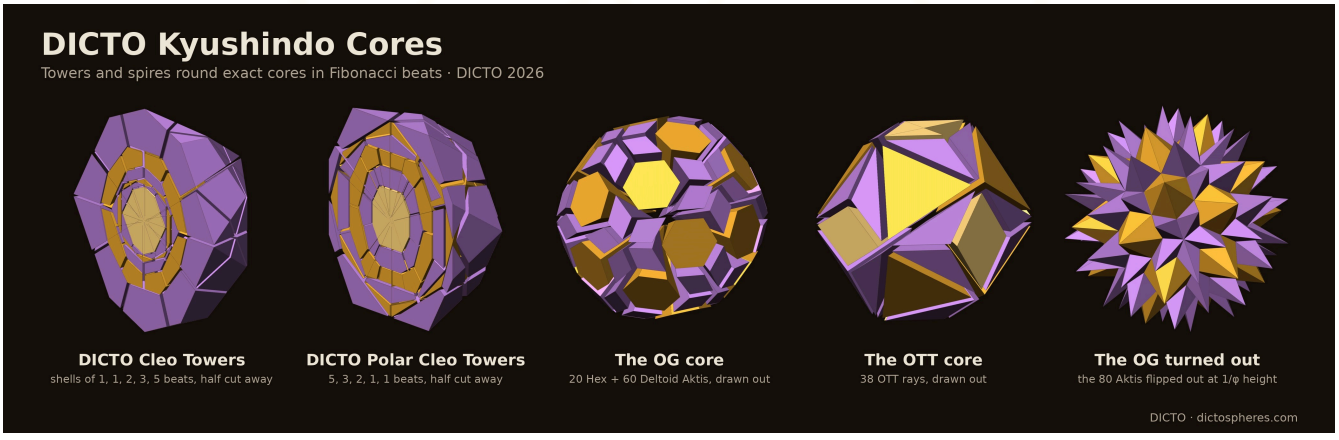
DICTO Kyushindo Cores

Towers and spires round exact cores in Fibonacci beats: the DICTO Cleo Towers, the DICTO Polar Cleo Towers, the OG core of 80 Aktis, the OTT core of 38 rays (Oktaktis, Triaktis and Tetraktis), and the OG turned out

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Version 3 · 12 November 2026 (Version 2: 8 November 2026; Version 1: 21 October 2026) · DICTO DOI: 10.5281/zenodo.23532013 · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/ee8ff206/studies (research branch, commit ee8ff206: folders 2026-11-09-og-margin, 2026-10-20-cleo-towers, 2026-10-20-og-spires, 2026-10-20-ott-rays and 2026-10-21-og-turned-out, with the exact tools of 2026-10-19-five-fold-octet and 2026-10-18-ocd-octet; also attached to this record)
The ideas (shells of towers in a Fibonacci rhythm round the CLEO sphere; their polar; the OG and OTT cut into rays from the centre; turning the OG's rays out with the golden step) and the names Cleo Towers, Polar Cleo Towers, Hex Aktis, Deltoid Aktis, OG, OTT, Oktaktis, Triaktis, Tetraktis, the OG turned out and the Kyushindo series are DICTO's; computation, exact proofs (including the proof of the main theorem) and drafting with Claude (Anthropic). The rule of each step is that of Universal PiTet [1], the three-dimensional antipedal construction; its reverse is Smarandache's face pedal polyhedron [2, 9].

Version 3 adds Theorem 5: Conjecture 1 (the OG's margin) holds for every $k \le 40$, exactly for $k \le 13$ and by certified ball arithmetic beyond, with equality only at $k = 3$; Lemma 4 (the next solid reaches every face plane); Proposition 6, which reduces Conjecture 1 for all k to a lower bound on the inradius; and Table 10.

Version 2 adds Theorem 4: Conjecture 2 is proved, every convex polytope with octahedral rotational symmetry lies inside 3·P(S) (sharp, cube and octahedron), by the moments of the octahedral orbits; and Corollary 7: the pyritohedral constant 3 and the tetrahedral constant 9 (sharp, tetrahedron).



Abstract. The main theorem: every convex polyhedron S with icosahedral rotational symmetry lies inside $(15 - 6\sqrt{5})\cdot P(S)$, and so inside $\varphi\cdot P(S)$, where P(S) is the hull of the foot points of S's face planes (Theorem 2, sharp; from a general containment bound, Theorem 1). Hence DICTO's **OG turned out**, which flips the 80 face pyramids (Aktis) of the DICTO Ogdoecontahedron (OG) outward at $1/\varphi$ of their height and takes the hull, beat after beat, is exactly φ^k times the k-th solid of the OG's downward family [2] at beat k and is **nested for every k** (Corollary 1), although the downward family itself is not: its first break, TQH → TQP, is caused exactly by the 30 ribbons of the TQH, trapezoid pairs whose

foot points fall outside their faces (Proposition 1). Reciprocation exchanges the two steps, polar(P(S)) = T(polar S) (Lemma 3), so the polar of the OG turned out is the Universal PiTet sequence of the DICTO Hecatodihedron, shrinking inward and nested for every k. Reciprocation about the sphere of radius² $15 - 5\sqrt{5}$ maps beat k of the Universal PiTet rule [1] to beat $5 - k$ (Theorem 3). With a lemma that every shell splits into towers, congruent within each symmetry orbit, these give, exactly in $\mathbb{Q}(\sqrt{5})$: the **DICTO Cleo Towers** (shells of 1, 1, 2, 3 and 5 beats round the CLEO sphere, with 32, 30, 32, 32 and 60 towers), the **DICTO Polar Cleo Towers** (5, 3, 2, 1 and 1 beats from the centre; 30, 32, 30, 60 and 32 towers), the towers of the OG turned out (80, 132 and 140 in 2, 3 and 3 shapes) and of its polar (80, 132 and 162 in 2, 3 and 4 shapes). We also give the **OG core of 80 Aktis** (20 **DICTO Hex Aktis** and 60 **DICTO Deltoid Aktis**, taking $(65 - 3\sqrt{5})/110$ and $(45 + 3\sqrt{5})/110$ of the OG) and the OTT core of 38 OTT rays (8 **DICTO Oktaktis**, each exactly a Cleo-Icos lying in a hexagon plane of the OG, 24 **DICTO Triaktis** and 6 **DICTO Tetraktis**). The boundary solids change their surface pattern from shell to shell; on the TQH the trapezoids pair into 30 ribbons on the two-fold axes, which DICTO sees as a possible new shell structure; they return exactly at down 8 and down 9; on the hulls the two-fold axis word, of period 4 from the OG to down 7, breaks at down 8 (the axis point is a listed foot point but not a hull corner), and the listed points themselves leave the period at down 10. Each result is marked as proved in general, proved for the cases built, or observed.

1. Definitions: beats, shells and towers

The beat. As in Universal PiTet [1]: for a convex solid S centred at the origin, $T(S) = \{x : v \cdot x \leq v \cdot v$ for every corner v of S$\}$. Each face of T(S) passes through a corner of S at right angles to the line from the centre. A corner v of S stays on the surface of T(S) exactly when $w \cdot v \leq w \cdot w$ for every corner w of S (otherwise the face of w cuts it off). This is not true of every convex solid, but it holds, exactly, for every station stepped from in this paper: the ID, RT, GTI and FPD, and so for all their scaled copies, since the condition does not change with scale (it holds too for the OG, TPH, TQH and TQHD, but not for the TQP, where 60 of the 140 corners would be cut off). In the plane this is the classical antipedal construction about a point; T is its three-dimensional form [1, 10], and the reverse step P, whose corners are the foot points of S's face planes (the points of the planes nearest the centre), is Smarandache's face pedal polyhedron [9], run from the OG in [2]. DICTO calls one step a beat.

The rhythm. DICTO's idea: shells of towers as tall as possible, in a Fibonacci rhythm. Shell n fuses F_n beats, 1, 1, 2, 3, 5, so the shells end at beats 0 | 1 | 2 | 4 | 7 | 12, and the beats inside a shell are passed through, not drawn.

The towers. A shell between an inner solid A and an outer solid $B \supset A$ is cut into one tower per face of A: the cone from the centre over that face, cut off by B, without the pyramid from the centre to the face (these are the shell pieces of [1]). By Lemma 1 (§2) the towers fill the shell and towers on faces of one symmetry orbit are congruent; for every shell below we also check exactly that the towers on each kind of face form one orbit of the 60 exact rotations of the icosahedral group, have one exact volume, and add up to the volume of the shell with remainder exactly 0, and that the inner and the outer solid of every shell are each fixed by all 60 rotations. Corners are proved as in [1]: floating point only guesses which faces meet at a corner; every corner then solves its faces exactly and lies inside every face, every face holds at least three corners, and every edge lies in exactly two faces with $V - E + F = 2$.

The OG turned out (DICTO's idea: turn the inside towers out and start a new sequence). Cut the DICTO Ogdoecontahedron (OG) [3] into its 80 face pyramids, the Aktis (§5), and flip each outward over its own face, by the **golden step**: each Aktis shortened to $1/\varphi$ of its height before it is flipped, so its tip lies at φ times the foot point of its face plane (the point of the plane nearest the centre). The hull of the 80 tips is the next solid S_1; on it the same is done again, one beat at a time: S_0 = OG, S_k = the hull of the tips of S_{k-1}. The OG's **downward family** [2] is D_0 = OG, $D_k = P(D_{k-1})$, the hull of the foot points of D_{k-1}'s face planes; D_1, …, D_4 are DICTO's TPH, TQH, TQP and TQHD (Triangle-Pentagon Hecatontatriacontadihedron, Triangle-Quadrilateral-Hexagon Polyhedron, Triangle-Quadrilateral-Pentagon Polyhedron and Triangle-Quadrilateral-Hexagon Diacosihedron), and we call the next ones down 5, down 6, …. DICTO named these by their face kinds and face count (Table 9): down 5 to down 11 are the DICTO TQP-252, TQ-320, TQP-402, TQ-500, TQD-732, TQHD-662 (not the TQHD of [2]) and TQP-672; below, down k and its name are used interchangeably.

What is proved. Each result is marked: *general*, proved for every k or every solid of the kind stated; *built*, proved exactly in $\mathbb{Q}(\sqrt{5})$ for every case built (the cases are listed); *observed*, seen in the cases built and not proved.

2. Main results

Notation: for a convex polytope S with the centre O inside, with faces $n_i \cdot x \le h_i$ (unit normals, $h_i > 0$), R(S) is the largest corner distance and r(S) the smallest face distance from O; $t_{min}(S)$ is the least t with $S \subset t \cdot P(S)$. By Theorem 1 below, $S_{k-1} \subset S_k$ in the OG turned out holds exactly when $t_{min}(D_{k-1}) \le \varphi$. Theorems 1 and 2 and their proofs were found with Claude (Anthropic) for DICTO's construction; the full proofs are in the study folder 2026-10-21-nesting-proof (README.md).

Theorem 1 (containment constant; *general*). For every convex polytope S with O inside, $\max_v |v|^2/h_P(v) \le t_{min}(S) \le R(S)/r(P(S)) \le (R(S)/r(S))^2$, where $h_P(v) = \max\{p \cdot v : p \in P(S)\}$ is the support function of P(S) at the corner v. Equality holds throughout for the five Platonic solids (9 for the tetrahedron, 3 for the cube and octahedron, $15 - 6\sqrt{5}$ for the dodecahedron and icosahedron). No constant works for all polytopes, not even centrally symmetric ones: thin square bipyramids have $t_{min} = (R/r)^2 \to \infty$.

Proof. For a unit vector u let $s \cdot u$ be the boundary point of S in direction u, on face i; then $s \le R$ and $n_i \cdot u = h_i/s \ge h_i/R$, so P(S), which contains the foot point $h_i n_i$, has support at least $h_i(n_i \cdot u) \ge r^2/R$ in every direction and so contains the ball of radius r^2/R; hence $r(P(S)) \ge r^2/R$, and S, inside the ball of radius R, lies in $(R/r(P(S))) \cdot P(S) \subset (R/r)^2 \cdot P(S)$. For the lower bound, P(S) lies in the half-space $v \cdot x \le h_P(v)$, so $v \in t \cdot P(S)$ needs $|v|^2 = v \cdot v \le t\, h_P(v)$. For a Platonic solid all $|v| = R$ and all $h_i = r$, and the face normal nearest a corner makes the angle $\arccos(r/R)$ with it, so both bounds equal $(R/r)^2$. ∎

Theorem 2 (icosahedral nesting; *general*). Every convex polyhedron S with icosahedral rotational symmetry about O satisfies $S \subset (15 - 6\sqrt{5}) \cdot P(S) \subset \varphi \cdot P(S)$; $15 - 6\sqrt{5} \approx 1.583592 < \varphi$, since $29 < 13\sqrt{5}$ ($841 < 845$), and the constant is sharp (icosahedron, dodecahedron).

Proof. By Theorem 1 it suffices that $r(S) \ge c \cdot R(S)$ with $c = \sqrt{((5 + 2\sqrt{5})/15)} \approx 0.794654$, the inradius-to-circumradius ratio of the icosahedron, since then $(R/r)^2 \le 1/c^2 = 15 - 6\sqrt{5}$. Let v

be a farthest corner. S is convex and symmetric, so it contains the hull of the orbit of v, and every face plane, with unit normal w, has distance at least $\max_g$ (g v)·w. So it is enough to show the **covering lemma**: for all unit u, w some rotation g of the icosahedral group I has (g u)·w ≥ c, i.e. every orbit comes within θ* = arccos c ≈ 37.377° of every direction. The reflection group I_h tiles the sphere by 120 copies of the (2, 3, 5) Möbius triangle T, with sides 20.905°, 31.717° and 37.377°; a spherical convex polygon with all vertex distances below 90° has diameter equal to its longest vertex distance, so diam T = θ*. The kite K = T ∪ sT (s the reflection in the longest side) has diagonals θ* and 2 × 18° = 36°, by the spherical sine rule (sin of the altitude = sin 20.905° sin 31.717°/sin 37.377° = sin 18° exactly), so diam K = θ* too. Move u into T by I_h (replacing the orbit by its antipodal image, which has the same covering radius) and pick h in I_h with h w in T: if h is a rotation, $d(h^{-1}u, w) \le \theta^*$; otherwise h = s g with g in I, and u and g w both lie in K, so $d(g^{-1}u, w) \le \theta^*$. ∎ (Checked numerically, including the 18° altitude and the minimum inradius of orbit hulls, in covering_lemma.py.)

Corollary 1 (the OG turned out; *general*). $S_k = \varphi^k \cdot D_k$ for every k, and the sequence is nested, $S_{k-1} \subset S_k$, for every k; the same holds for the golden tower $\varphi^k P^k(S)$ of every icosahedrally symmetric start S. The downward family itself does not nest: $D_k \subset D_{k-1}$ holds only for k = 1, 2 and 4 among k ≤ 9 (Proposition 1: the feet of D_8's 120 outlying triangles put D_9 outside D_8).

Proof. $S_k = \varphi^k D_k$ by induction: $S_0 = D_0$; if $S_{k-1} = \varphi^{k-1} D_{k-1}$, its face planes and their foot points are those of D_{k-1} scaled by φ^{k-1}, the tips are φ times these, so $S_k = \varphi^k P(D_{k-1}) = \varphi^k D_k$ (P commutes with scaling). Every D_k is icosahedrally symmetric, so by Theorem 2 $D_{k-1} \subset \varphi \cdot P(D_{k-1}) = \varphi \cdot D_k$; multiply by φ^{k-1}. ∎ The exact margins $t_{\min}(D_k)$ for k = 0, …, 8 are in Table 7 (all ≤ $3/\varphi^2 \approx 1.1459$, with equality at k = 3); the nesting was also checked directly in $\mathbb{Q}(\sqrt{5})$ for k ≤ 8 (nest_feet.py).

Theorem 3 (polarity of the beats; *general*). Let S_k be the beats of the Universal PiTet rule from the icosidodecahedron of circumradius φ (S_0 = ID, S_1 = RT, S_2 = GTI, S_3 = FPD, $S_{k+4} = c \cdot S_k$, $c = 10/\varphi^4$ [1]), for all integers k. Reciprocation about the sphere of radius² $\rho^2 = 15 - 5\sqrt{5} = c \cdot \varphi^2$ maps S_k to S_{5-k} for every k.

Proof. For k = 0, …, 5 the polar $\{x : v \cdot x \le \rho^2$ for every corner v of $S_k\}$ is computed exactly and equals S_{5-k}. Polarity reverses scaling, polar(c·S) = polar(S)/c, so $\text{polar}(S_{k+4}) = S_{5-k}/c = S_{1-k}$, and the six cases give every k. This is the self-polarity of [1, Proposition (iv)], read beat by beat. ∎

Lemma 3 (polarity exchanges the two steps; *general*; elementary, but not found in the sources checked). For a convex polytope S with O inside and reciprocation about any sphere centred at O, polar(P(S)) = T(polar(S)).

Proof. Let S have faces $n_i \cdot x \le d_i$. Then polar(S) is the hull of the points $w_i = \rho^2 n_i / d_i$, so $T(\text{polar}(S)) = \{x : w_i \cdot x \le w_i \cdot w_i\} = \{x : n_i \cdot x \le \rho^2 |n_i|^2 / d_i\}$. P(S) is the hull of the foot points $f_i = d_i n_i / |n_i|^2$, so $\text{polar}(P(S)) = \{x : f_i \cdot x \le \rho^2\} = \{x : n_i \cdot x \le \rho^2 |n_i|^2 / d_i\}$, the same set. ∎

Corollary 2 (the polar of the OG turned out; *general*). polar(S_k) = $\varphi^{-k}\cdot T^k$(polar OG) = $\varphi^{-k}\cdot T^k$(Hecatodihedron) (the OG is the polar of the DICTO Hecatodihedron [3]; about the unit sphere, polar(OG) is a scaled Hecatodihedron, its counts and face kinds checked exactly), and these solids nest inward for every k, polar(S_k) ⊂ polar(S_{k-1}), since polarity reverses inclusion (Corollary 1). (*Built:* checked exactly for k ≤ 4, §7.)

Proof. Induction with Lemma 3 and polar(φX) = polar(X)/φ: polar(S_k) = φ^{-k}polar(P(D_{k-1})) = φ^{-k}T(polar(D_{k-1})) = … = $\varphi^{-k}T^k$(polar(OG)). ∎

Lemma 1 (towers; *general*). Let A ⊂ B be convex solids with the centre inside A, both fixed by a rotation group G. The towers over the faces of A fill B \ A and meet only in their boundaries, and towers over faces in one G-orbit are congruent.

Proof. The cones from an inside point over the faces of A cover space and meet only in boundaries; cut by B they cover B, and removing the pyramids from the centre over the faces, which make up A, leaves B \ A. A rotation g in G with g(F) = F′ maps the cone over F onto the cone over F′ and fixes A and B, so it maps the tower on F onto the tower on F′. ∎

Lemma 2 (foot points; *general*). For a convex solid S with the centre inside it, P(S) ⊂ S exactly when the foot point of every face plane lies in its own (closed) face; a foot point outside its face lies outside S.

Proof. A face plane supports S, and S meets it exactly in the face; the foot point lies on the plane, so it is in S exactly when it is in the face. P(S) is the hull of the foot points, and lies in the convex S exactly when all of them do. ∎

Proposition 1 (the ribbons and the break; *built*). On the OG and the TPH every foot point lies inside its face, so TPH ⊂ OG and TQH ⊂ TPH. On the TQH the foot points of the 60 trapezoids, the 30 ribbons of §8, lie outside their faces and all others inside; by Lemma 2 the TQP is not inside the TQH, and its 60 corners from those feet are the ones standing outside, as found in [2]. On the TQP the feet of the 60 triangles lie on the edges of their faces and all others inside, so TQHD ⊂ TQP. Further down the feet of other faces fall outside (the 60 trapezoids of the TQHD, which lie in rings of five, 60 four-sided faces of down 5, 180 triangles of down 6, 240 of down 7 and 120 of down 8), and the downward family does not nest again. So the first break of the downward family is caused exactly by the ribbons; turned out, the factor φ per beat absorbs every such break (Corollary 1). ∎ (exact: nest_feet.py)

Remark (the reverse step is not always undone). T(P(S)) = S holds when every foot point of S is a corner of P(S). For D_6, D_7 and D_8 the reverse step fails: 60, 60 and 90 of the listed foot points of D_5, D_6 and D_7 lie strictly inside the next hull, whose true corners number 192 of 252 (D_6), 260 of 320 (D_7) and 312 of 402 (D_8); at D_9 it holds again, all 500 listed points being corners. The tower and Corollary 1 use only the hulls and are unaffected. Face counts are not monotone either: the points listed for D_{k+1} are the faces of D_k, and D_{10} has 662 faces against 732 for D_9.

3. The DICTO Cleo Towers (Corollary 3)

The core is DICTO's CLEO sphere [7, §5.5]: the icosidodecahedron (ID), cut from its centre into 20 CLEOs [6] on its triangles and 12 pentagonal pockets. Here it has circumradius φ

(edge 1), the frame of [1]. Its beats are the stations of Universal PiTet: S_0 = ID, S_1 = RT (rhombic triacontahedron), S_2 = DICTO GTI (Golden Truncated Icosahedron), S_3 = DICTO FPD (Flat Pentakis Dodecahedron), and $S_4 = c \cdot ID$ with $c = 10/\varphi^4 = 35 - 15\sqrt{5} \approx 1.459$: one turn of four beats returns the ID exactly c times larger (checked again here), so $S_{k+4} = c \cdot S_k$ for every k [1].

So the five shells hold 32, 30, 32, 32 and 60 towers, 186 in all, and the solid of 12 beats is the ID again, $c^3 \approx 3.11$ times as wide as the core. Shells 1 and 2 are the first two shells of [1]; shells 3, 4 and 5, which fuse beats, are new. In each fused shell the towers stand on the faces of the shell's inner solid only, so the solids passed through leave no trace: shell 3, for instance, has towers on the GTI's 20 hexagons and 12 pentagons that reach to the corners of c·ID.



Figure 1. The DICTO Cleo Towers. The core (cream) and each shell's towers on their inner solid, drawn slightly apart, the shells alternately purple and gold; last, all five shells together with half cut away, the towers drawn apart. The labels c and c^3 are $10/\varphi^4$ and its cube.

4. The DICTO Polar Cleo Towers (Corollary 4)

The polar of the Fibonacci shells. Take the sphere through the GTI's corners, radius² $\rho^2 = 15 - 5\sqrt{5} = c \cdot \varphi^2$: $10/\varphi^4$ times the CLEO sphere's radius². Polar reciprocation about it takes every beat to another beat: **$\text{polar}(S_k) = S_{5-k}$**, checked exactly for k = 0, …, 5 (ID ↔ c·RT, RT ↔ c·ID, GTI ↔ FPD; this is the self-polarity of [1, Proposition (iv)]); since polar(c·S) = polar(S)/c, it holds for every k (Theorem 3). The boundaries 0 | 1 | 2 | 4 | 7 | 12 of the Cleo Towers go to 5 | 4 | 3 | 1 | −2 | −7: read from the centre, shells of **5, 3, 2, 1 and 1 beats**, the most beats deepest. The core is $S_{-7} = RT/c^2$, which cuts from its centre into 30 rhombic rays.

The polar shells hold 30, 32, 30, 60 and 32 towers, 184 in all. Shell 4 is the last shell of [1]. Shell 5 is the polar image of the Cleo Towers' first shell and, because polarity swaps the ID and the RT, it is that same shell c times larger: its towers are those of the Cleo Towers' shell 1 (Table 1, Appendix A), scaled by c (volumes by c^3).



Figure 2. The DICTO Polar Cleo Towers. The core, the rhombic triacontahedron RT/c² cut into 30 rhombic rays (cream), and each shell's towers on their inner solid, alternately purple and gold; last, all five shells with half cut away. "RT x c^-2" is RT/c², and so on.

5. The OG core of 80 Aktis (Proposition 2)

DICTO's idea: cut the DICTO Ogdoecontahedron (OG) [3], with its 20 regular hexagons and 60 kites, into its 80 face pyramids, every tip at the centre, as the CLEO sphere is cut into CLEOs and pockets. DICTO named them **Aktis** (Greek, ray): 20 **DICTO Hex Aktis** on the hexagons and 60 **DICTO Deltoid Aktis** on the kites. Values are in the frame of [4], where the OG has 102 corners and volume $(-1000 + 1240\sqrt{5})/1539 \approx 1.151868$.

- The 80 Aktis fill the OG exactly (volumes add up with remainder 0; neighbouring Aktis meet only in faces, as pyramids over different faces of a convex solid from an inside point must; the first 60 of the 180 neighbouring pairs were also intersected exactly).
- The 20 Hex Aktis are one shape, and the 60 Deltoid Aktis are one shape (corner matching: tip to tip, base corners in order, all distances equal).
- **Hex Aktis:** volume $(-2 + 2\sqrt{5})/81 \approx 0.030520$, height $\sqrt{((6 + 2\sqrt{5})/27)} \approx 0.622782$; the 20 take $(65 - 3\sqrt{5})/110 \approx 52.99\%$ of the OG.
- **Deltoid Aktis:** volume $(-4 + 8\sqrt{5})/1539 \approx 0.009024$, height $\sqrt{((42 + 2\sqrt{5})/109)} \approx 0.652955$; the 60 take $(45 + 3\sqrt{5})/110 \approx 47.01\%$.

The kites lie farther out than the hexagons, so the Deltoid Aktis are the taller rays.

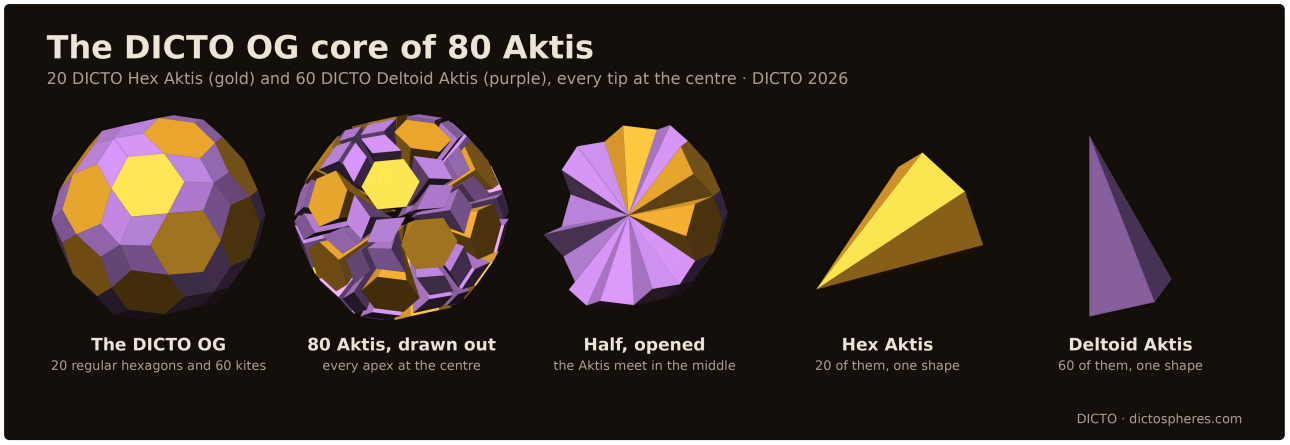


Figure 3. The OG core: the DICTO OG; its 80 Aktis drawn out, every tip at the centre; half of them, opened; one Hex Aktis (gold) and one Deltoid Aktis (purple).

6. The OTT core of 38 rays (Proposition 3)

The DICTO OTT (Octahedrally Truncated Triacontahedron) [5] is the rhombic triacontahedron cut by its octahedron at all 8 key sites: 24 corners, 38 faces (8 equilateral triangles in the octahedron's faces, 24 half-rhombi and 6 whole rhombi), volume $(-608 + 320\sqrt{5})/81 \approx 1.327676$, pyritohedral symmetry. DICTO's idea: cut it, like the OG, into its face pyramids from the centre, the **OTT rays**, which DICTO named **Oktaktis** (on the octahedron cuts), **Triaktis** (on the half-rhombi) and **Tetraktis** (four-sided, on the whole rhombi). The 38 rays fill the OTT exactly and fall into three shapes (corner matching); their values are in Table 3.

The Oktaktis are Cleo-Icos. DICTO noticed that the Oktaktis looks like a Cleo-Icos; we prove it exactly: each Oktaktis is exactly similar to the Cleo-Icos [7, §5.6], the pyramid from an icosahedron's centre to one face: $\text{height}^2/\text{edge}^2 = \varphi^4/12$ and $\text{side}^2/\text{edge}^2 = (\varphi^2 + 1)/4$, as for the icosahedron. **They lie in the OG's hexagon planes.** In the frame of [4], where both solids are given, each of the OTT's 8 octahedral triangles lies exactly in one of the 20 hexagon planes of the OG; so an Oktaktis has exactly the height of a Hex Aktis, $\sqrt{((6 + 2\sqrt{5})/27)}$. Since the OG lies inside the OTT [5], each of those 8 hexagons of the OG lies in an octahedral triangle of the OTT; more exactly, the hexagon's circumradius equals the triangle's inradius (squared, $(6 - 2\sqrt{5})/27$), and three corners of the hexagon are exactly the midpoints of the triangle's edges, so each hexagon touches its triangle at those three points.



Figure 4. The OTT core: the DICTO OTT; its 38 rays drawn out, every tip at the centre; half of them, opened; one Oktaktis (gold), one Triaktis (purple) and one Tetraktis (cream).

7. The OG turned out and its polar: the shells (Corollaries 5 and 6)

The beats. By Corollary 1 the beats of the OG turned out are the OG, φ·TPH, φ²·TQH, φ³·TQP, φ⁴·TQHD, …; at each of the first four beats the corners are checked to be exactly the flipped tips, and each solid is fixed by the 60 rotations of the icosahedron (Table 4a). Every beat holds the one before (Corollary 1), also from φ²·TQH to φ³·TQP, where the downward family itself breaks (Proposition 1). Beat k is exactly φ^k times the k-th downward solid, but the downward solids themselves shrink [2], so the outer radius grows by a little less than φ per beat: 1.52, 1.53, 1.61 and 1.51.

Towers (Corollary 5, *built*). The shells of 1, 1 and 2 beats are cut into towers by Lemma 1; their exact values are in Table 4b.

So the shells hold **80, 132 and 140 towers** of **2, 3 and 3 shapes**. In shell 2 the 120 triangles of the TPH carry towers of two volumes; split by the 60 rotations they form two orbits of 60, each of one volume, so they are two shapes (60 + 60), and with the 12 pentagon towers the shell has three.

The full flip (float, for comparison). Flipping the Aktis at their whole height puts each tip at 2 times the foot point. The same argument gives beat k = 2^k times the k-th downward solid, and a float run agrees: 102, 80, 132, 140 and 162 corners at beats 0–4, the outer radius growing by about 1.9 per beat (flip-float.json). The golden step gives the same solids, growing by a little less than φ.

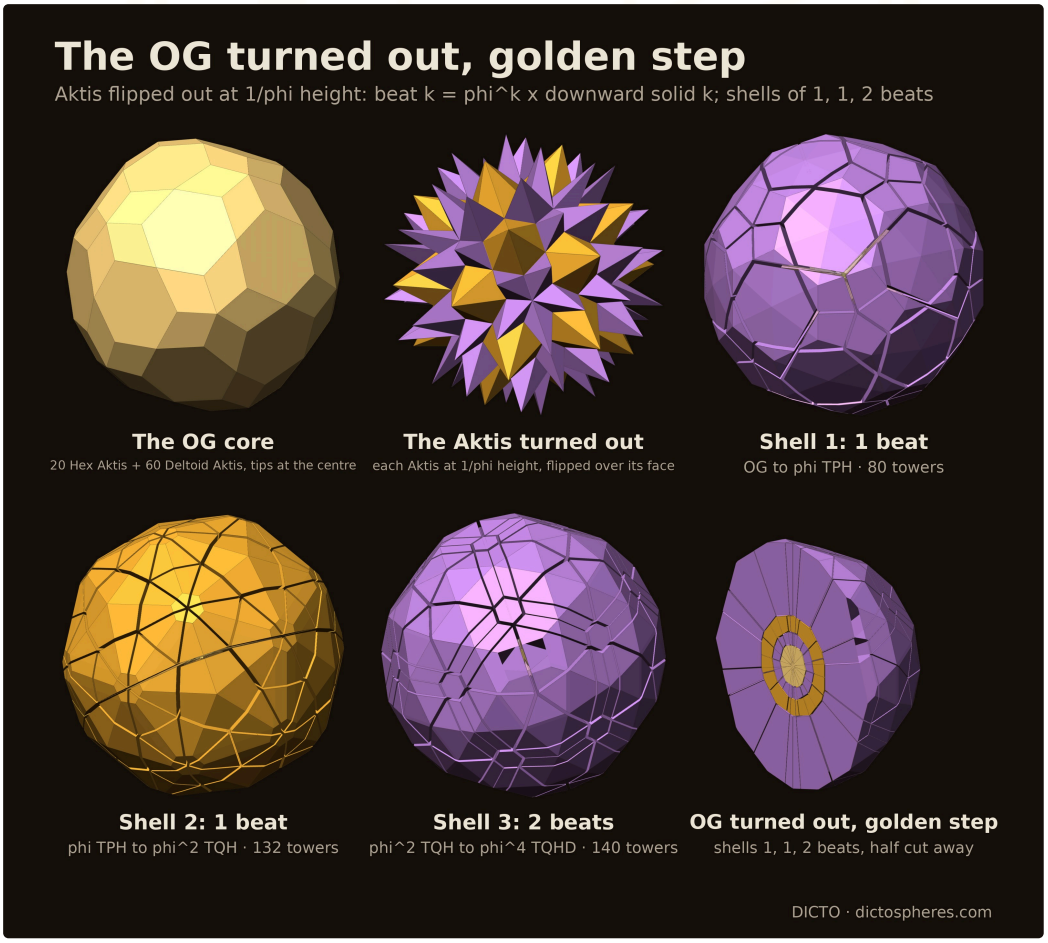


Figure 5. The OG turned out, golden step. The OG core (cream); the 80 Aktis at 1/φ of their height, flipped over their faces (Hex Aktis gold, Deltoid Aktis purple); shells 1, 2 and 3 (1, 1 and 2 beats) on their inner solids, purple, gold and purple; all three with half cut away. "phi TPH" is φ·TPH, and so on.

Shell 4 (3 beats, beat 4 → beat 7; *built*). Exact: both boundaries, φ^4·TQHD and φ^7·TQP-402, are fixed by all 60 rotations, beat 4 lies inside beat 7, and the shell is cut into **200 towers** on the TQHD's faces: 120 on triangles with two volumes, 60 of each (≈ 570.644662 and 889.860165), 60 on quadrilaterals (≈ 147.140716) and 20 on hexagons (≈ 5699.759901), adding up to the shell (≈ 210453.9306, frame of [2]) with remainder exactly 0 (shell4_exact.py; its one FAIL line is the test "one volume per kind of face", the triangles having two). **Shell 5** (5 beats, beat 7 → beat 12) is not built: beat 12 (down 12) has not been computed; its nesting is covered by Theorem 2. Down 5 to down 10 are built exactly; of down 11 only the exact points are built (its faces from float tests).

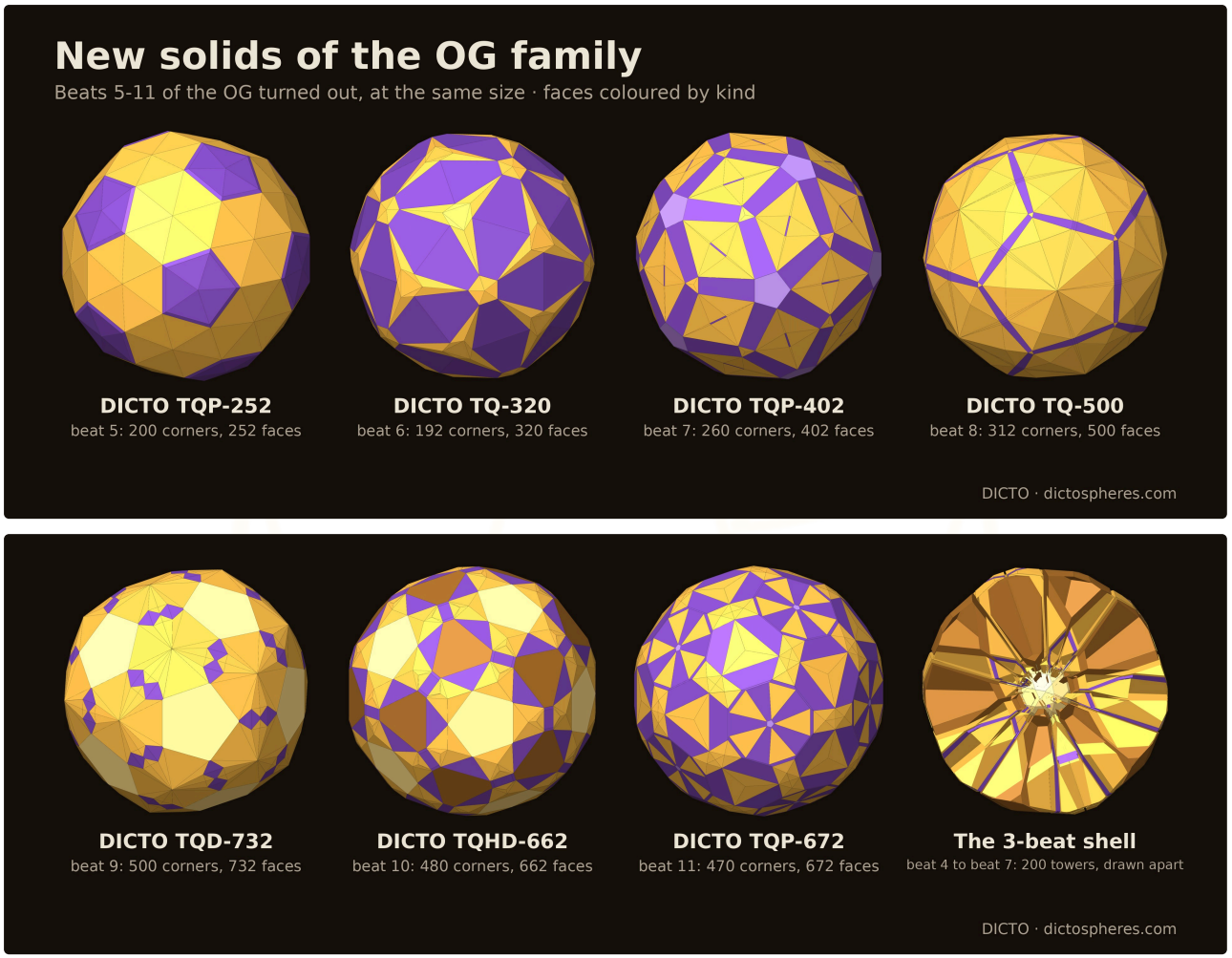


Figure 6. The new solids of the OG's downward family, beats 5 to 11 of the OG turned out, each to the same size, named by DICTO (face kinds measured on the exact points by float tests; triangles gold, four-sided faces purple, pentagons lilac, hexagons bronze, decagons pale); last, the 3-beat shell (beat 4 to beat 7), its 200 towers drawn apart. Every tower shape of every shell is shown in Figure 11 (Appendix A).

The polar sequence (Corollaries 2 and 6; DICTO's idea: the polar cycles of the OG turned out). By Lemma 3, reciprocation turns the turned-out sequence into the UP steps from the DICTO Hecatodihedron, shrinking inward: polar(S_k) = $\varphi^{-k}T^k$(Hecatodihedron). Built exactly for $k \le 4$ (reciprocation about the unit sphere in the frame of [2]), the corner sets agree exactly; the solids have 80, 132, 140, 162 and 200 corners and 102, 80, 132, 140 and 162 faces (polar S_0, the Hecatodihedron: 60 3-gons, 30 4-gons, 12 5-gons; polar S_4: 90 four-sided faces in two kinds, 30 and 60, and 72 pentagons in two kinds, 60 and 12), and each lies inside the one before. **Corollary 6** (*built*): the inward shells of 1, 1 and 2 beats, cut into towers on

the faces of each shell's inner solid by Lemma 1, hold 80, 132 and 162 towers of 2, 3 and 4 shapes, each shell filled exactly (Table 8).



Figure 7. The polar of the OG turned out: the DICTO Hecatodihedron, scaled (cream), the outermost boundary; the inward shells of 1, 1 and 2 beats, each shell's towers on its inner solid (pale), purple, gold and purple; all three with half cut away. "polar S1" is polar(S_1), and so on.

8. The surface patterns, the lines and the axes (Propositions 4 and 5)

DICTO observed that every shell so far has a different pattern on its surface, and that "the lines defining the shells denote the structures". We read each boundary solid two ways, both exactly: by its faces (the orbits of the 60 rotations, and how faces of one kind join edge to edge into groups) and by its lines (the edge network). The boundaries are few solids: the Cleo Towers end at ID | RT | GTI | ID | FPD | ID (each $10/\varphi^4$ to a power larger), the Polar Cleo Towers at RT | GTI | RT | FPD | ID | RT, and the OG turned out at OG | TPH | TQH | TQHD (each φ to a power larger; the TQP is passed through inside shell 3). Scale does not change a pattern, so Table 5 (Appendix A) lists the nine solids once each.

Proposition 4 (surface patterns; *built*: the nine solids of Table 5). The faces and lines of the boundary solids are as in Table 5; in particular:

Faces. No two boundaries of one sequence in a row have the same faces. In the OG's sequence each boundary solid has its own grouping: kites in fives (OG), triangle pairs on the two-fold axes (TPH), trapezoid ribbons on the two-fold axes (TQH), and trapezoid rings of five on the five-fold axes (TQHD), which repeat the rings of the TQP passed through inside shell 3.

Lines. No edge runs on straight through a corner, as convexity requires (a corner never lies between two points of the solid). The lines that line up across the surface are the parallel families. On every solid but the RT and the OG, 15 families point along the two-fold axes (on the RT, a zonohedron, the 6 families point along the five-fold axes); the OG alone has no edge along any axis. Only the ID has central belts: its six decagonal equators, as is classical.

The ribbons (DICTO's observation). On the TQH of the OG turned out (beat 2, φ^2·TQH) the 60 four-sided faces are congruent isosceles trapezoids, with parallel sides of squared length $(38367 + 17019\sqrt{5})/13448$ and $(6615 + 2925\sqrt{5})/3362$ (≈ 2.3839 and 1.9781), legs ≈ 0.4343, corner angles $\approx 62.15°$ and $117.85°$ and height ≈ 0.3840. They pair up across their shorter parallel sides: 30 shared edges, every trapezoid in exactly one pair, each pair centred on a two-fold axis (a half-turn about it swaps the two), two pairs per axis, one at each end. So each ribbon is drawn by three parallel lines, its two outer sides and the seam, bent $\approx 23.28°$ across the seam (float). The 15 edge families of the TQH that point along a two-fold axis are exactly

these lines: 6 per family, the two ribbons at the two ends of one two-fold axis, all running along a second two-fold axis. Shell 3 stands a long blade-like tower on each trapezoid, so each ribbon carries two blades (Figure 8). In the TQP and the TQHD the trapezoids join along their legs instead, into rings of five on the five-fold axes (each trapezoid meeting two others); of the nine solids of Table 5 only the TQH has ribbons; further down they return (Proposition 5). DICTO sees in the ribbons a possible new shell structure; we record the observation and claim nothing beyond the checks above.

Proposition 5 (the axes; *built*: the OG to down 9). A symmetry axis leaves a convex solid with this symmetry through a corner on the axis, a face square to the axis or (on a two-fold axis only) the inside of an edge. For every solid of the downward family from the OG to down 9, which of the three it is, on each kind of axis, is as in Table 6 (exact: zero cross products in $\mathbb{Q}(\sqrt{5})$, on the corners and face planes of each solid's hull; from down 6 on, not every foot point of the reverse step is a corner of the hull: 192 of 252 points at down 6, 260 of 320 at down 7, 312 of 402 at down 8, and all 500 at down 9). At the two-fold edges of the TPH and down 5 two congruent isosceles triangles meet; at those of the TQH and down 8, a ribbon (two congruent isosceles trapezoids across their shorter parallel sides, the trapezoid's orbit having 60 faces, so 30 ribbons); at down 6, two congruent isosceles trapezoids across their longer parallel sides; at down 9, a ribbon again. So the ribbons of the TQH return exactly at down 8 and at down 9.



Figure 8. The ribbons. Left: the TQH (beat 2 of the OG turned out), triangles purple, hexagons cream, the 30 ribbons of trapezoid pairs coral. Right: one ribbon with its two shell-3 towers (coral, drawn a little apart across the seam), on the TQH's nearby faces (cream).

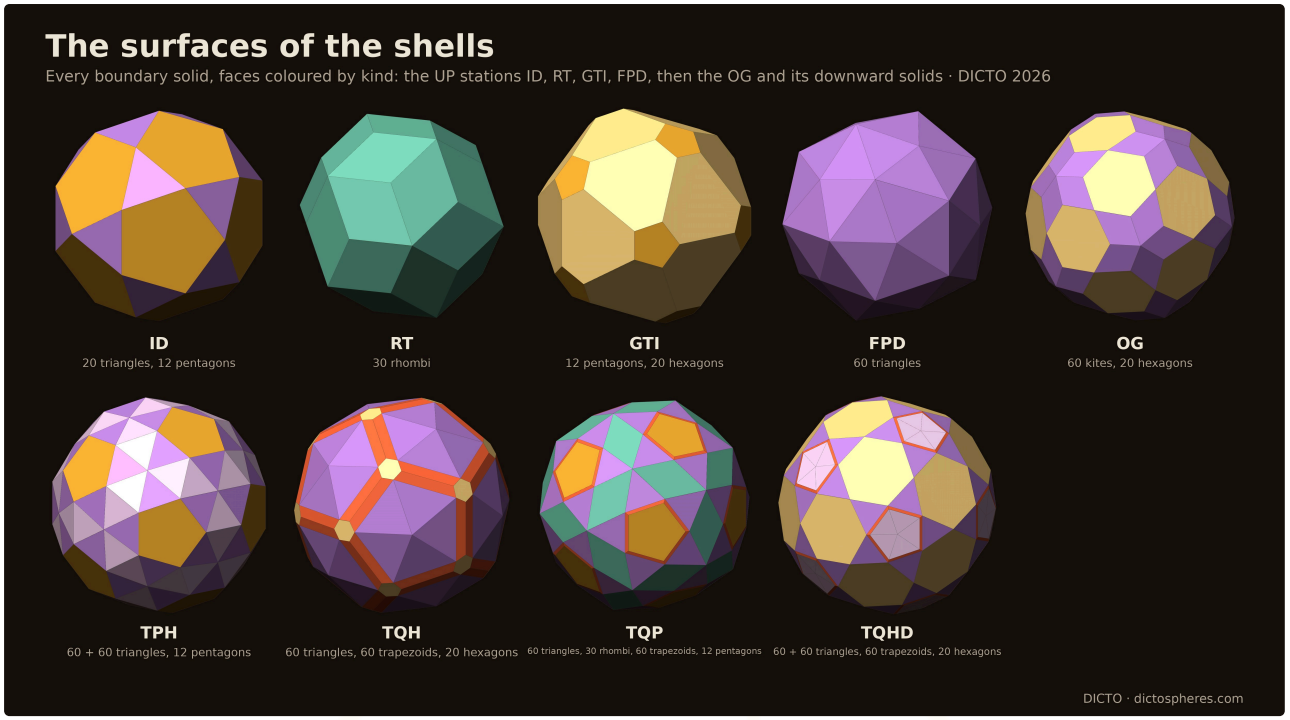


Figure 9. The surfaces of the shells: the nine boundary solids, all seen the same way, each to its own size. (The polar sequence and the axis rhythm: Figure 12, Appendix A.) Faces coloured by kind: triangles purple (a second kind of triangle lilac), pentagons gold, hexagons cream, kites purple, rhombi teal, trapezoids coral.

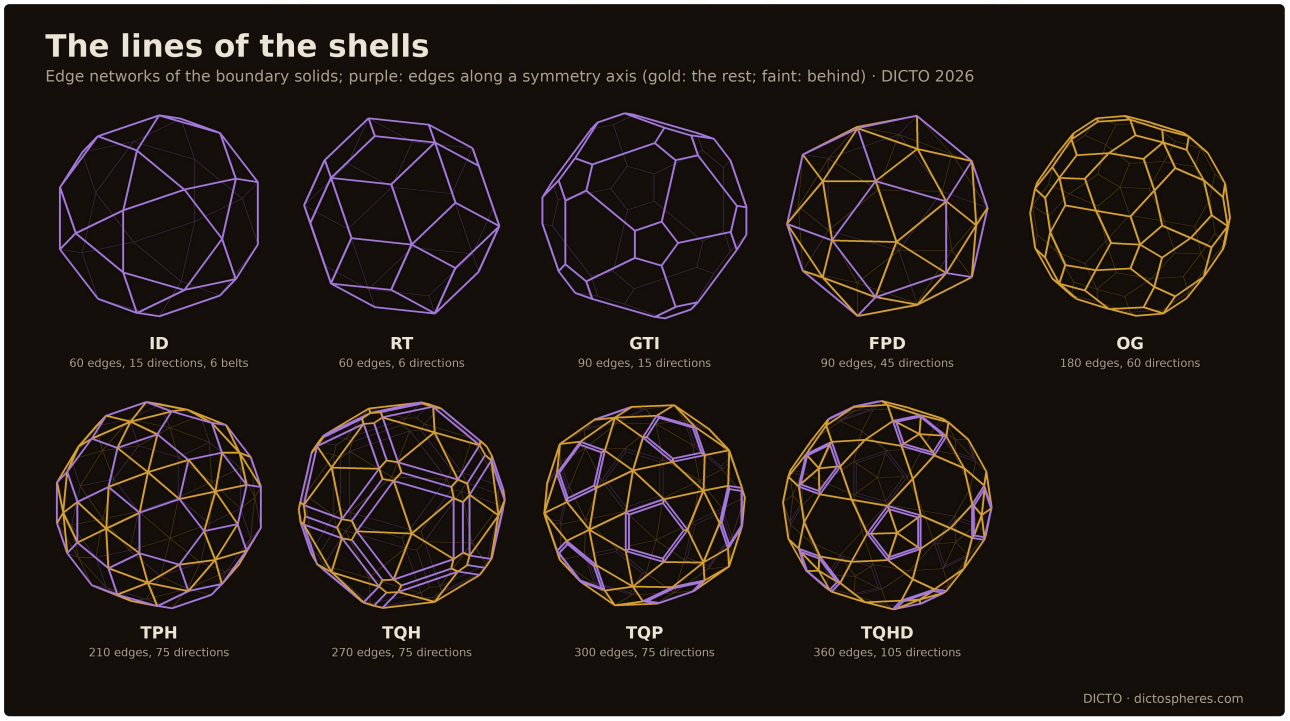


Figure 10. The lines of the shells: the edge network of each boundary solid, seen as in Figure 9. Purple: the edges that point along a symmetry axis (on the ID, the RT and the GTI all of them); gold: the rest; faint: the lines behind.

9. The OG's margin, the octahedral constant and an observation

Conjecture 1 (the OG's margin). $t_{min}(D_k) \le 3/\varphi^2$ for every k. *Evidence:* proved for $k \le 40$ (Theorem 5); exact for k = 0, …, 8 (Table 7), with equality at k = 3; the values are not monotone (1.134, 1.117, 1.103, 1.146, 1.042, 1.087, 1.026, 1.060, 1.138). Corollary 1 does not need it.

Theorem 5 (the OG's margin to k = 40; *exact for k ≤ 13, certified for 14 ≤ k ≤ 40*). For every k = 0, 1, …, 40, $t_{min}(D_k) \le 3/\varphi^2$, with equality only at k = 3, where $t_{min}(D_3) = (R/H)^2(D_3) = (R/r)^2(D_4) = 3/\varphi^2 = (9 - 3\sqrt{5})/2$ exactly: the farthest corner of D_3 and the nearest face of D_4 lie on the same axis. The margin then shrinks towards 1 without being monotone: $t_{min}(D_{40}) \le 1.000108$ (Table 10).

Proof. For k ≤ 13 the solids D_0, …, D_{14} were built in $\mathbb{Q}(\sqrt{5})$ and t_{min} computed exactly (Table 7 for k ≤ 8, Table 10 for 9 ≤ k ≤ 13). From k = 10 on, a certified computation carries the family: points are kept as interval balls (600 bits) for one foot point per symmetry orbit, every other point being the image of its orbit point under the exact matrices of I_h. Each face orbit is certified: three member points with certified non-zero area give the plane; every other point is proved strictly inside it (in floating point with a proved error bound of 10^{-12} where the margin is larger than 10^{-10}, in ball arithmetic otherwise); a face with more than three corners is accepted only when its coplanarity is forced by symmetry (one orbit of its stabiliser, or a mirror pairing), never by a numerical equality; and the face set is closed (every polygon edge in exactly two faces, the set invariant under I_h), so it is the whole boundary. Then $t_{min}(D_{k-1})$ = max over points v of D_{k-1} and faces (m, d) of D_k of m·v/d is bounded above with the same error bound and compared with $3/\varphi^2$. The values for 10 ≤ k ≤ 13 agree with the exact ones, and those for k ≤ 30 with an independent earlier certified run, to 10^{-9}. ∎

Lemma 4 (the next solid reaches every face plane; *general*). For every convex polytope S with O inside and every face normal n_i of S, $h_{P(S)}(n_i) \ge h_i$, with equality exactly when P(S) lies in that face's half-space. Hence a face of S that is also a face of P(S) never moves inward, and r(P(S)) ≥ min(r(S), ν), where ν is the least distance of a face of P(S) whose normal is not a face normal of S. *Proof.* The foot point $h_i n_i$ is a point of P(S), and its projection on n_i is h_i. ∎

Proposition 6 (the remaining step; *general reduction*). If $r(D_j) \ge (\varphi^2/3)\cdot R(D_{40})$ for every j > 40, then Conjecture 1 holds for every k. *Proof.* R never increases along the family (R(P(S)) = H(S) ≤ R(S)), and $t_{min}(D_k) \le R(D_k)/r(D_{k+1})$; Theorem 5 covers k ≤ 40. ∎ By Lemma 4 the inradius can fall only on new faces, and a new face spanned by foot points of radius ≥ r within angular radius γ lies at distance ≥ r cos γ; so the hypothesis follows if the new faces keep getting finer, $\Pi_{j>40} \cos \gamma_j \ge (\varphi^2/3)\cdot R(D_{40})/r(D_{40})$. In the computed range the largest new face shrinks by about 0.94 per level, and the margin is wide; a proof that it does so for every level is open (study 2026-11-09-og-margin, where every route tried, failed or open, is recorded).

Theorem 4 (octahedral nesting; *general*; Conjecture 2 of Version 1). Every convex polytope S with octahedral rotational symmetry about O satisfies S ⊂ 3·P(S), and the constant is sharp (cube, octahedron).

Proof. By Theorem 1 it suffices that r(S) ≥ R(S)/√3, and for that it suffices that for all unit vectors u, w some rotation g of the octahedral group O (24 rotations) has (gu)·w ≥ 1/√3: take u along a farthest corner and w along a face normal; the corner gRu of S then shows h ≥ R/√3 for that face. Let t_g = (gu)·w. Over the 24 rotations, the mean of t_g is 0 (no nonzero vector is fixed by O), the mean of t_g^2 is 1/3 (O acts irreducibly) and the mean of t_g^3 is 0 (there is no O-invariant cubic form: a quarter-turn changes the sign of xyz, the only invariant cubic of the tetrahedral group); the orbits are spherical 3-designs [27, 28]. Suppose every $t_g \le m$ with 0 < m < 1. The cubic $f(t) = (t - m)(t + 1/(3m))^2$ is ≤ 0 for t ≤ m, so its mean over the orbit is ≤ 0;

the moments give that mean as $(1 - 3m^2)/(9m)$. Hence $m \geq 1/\sqrt{3}$. Equality needs every t_g to be m or $-1/(3m)$, which happens for a four-fold axis against a three-fold axis: the cube and the octahedron, where Theorem 1 gives exactly 3. ∎ (A second proof repairs the kite of Theorem 2: averaging the three mirror images of the orbit triangle, not one, gives the bound; it also proves Theorem 2 again. Both are in the study folder 2026-11-08-octahedral-constant, every step checked in exact arithmetic.)

Corollary 7 (the other cubic groups; *general*). With pyritohedral symmetry (T_h, whose orbits are also 3-designs) $S \subset 3 \cdot P(S)$; with tetrahedral symmetry (T or T_d, whose orbits are only 2-designs) $S \subset 9 \cdot P(S)$, sharp for the tetrahedron. *Proof.* As above, with T_h's moments; for T, if every $t_g \leq m$ then $(m - t_g)(1 + t_g) \geq 0$, and its mean, by the first two moments, is $m - 1/3$; so $m \geq 1/3$ and $(R/r)^2 \leq 9$. ∎

Observation 1 (the axis rhythm; *observed*, Table 6). On the five- and three-fold axes corner and face alternate at every step from the OG to down 9 (period 2); one reason is plain: a face square to an axis has its foot point on the axis, and the reverse step makes it a point of the next solid (not always a corner, see Proposition 5). On the two-fold axes the word runs corner, edge, edge, face for two full periods, the OG to the TQP and the TQHD to down 7; at down 8 it reads edge where the period would give corner, the foot point on the axis not being a corner of the hull, and there the ribbons of the TQH return; at down 9 it reads edge again, as the period gives, with a ribbon there too, and at down 10 it reads face where the period gives edge (axis_words.py: no listed point of down 10 lies on the two-fold axis, and a face of its hull is square to it). So the edge stations and the ribbons recur in the solids built, but on the hulls the period 4 holds only from the OG to down 7: the hull word breaks at down 8 (the axis point is a listed foot point but not a hull corner), and the listed points themselves leave the period at down 10. For DICTO's reading of the ribbons as a possible shell structure, this is the law found so far; we claim nothing for solids not built.

10. Prior art and what is claimed

- **Known and credited.** The step T is the three-dimensional antipedal construction about the centre, and its reverse P is Smarandache's face pedal polyhedron [9, 10] (as recorded in [1, 2]); reciprocation about a sphere is classical [17]. Solids nested round one centre go back to Kepler's *Mysterium cosmographicum* [11]; icosahedral shells round a centre are found in Mackay's packing of equal spheres [12], in the clusters of icosahedral quasicrystals [13] and in the nested icosahedral point sets of viruses and carbon onions [14, 15]; Bergman's continuous spherical truss surrounds a core of twenty tetrahedral pyramids with endless layers [16]; Kramer filled space round a centre with cells that recur τ^3 larger [18]. Cutting a solid into pyramids from its centre is as old as Hypsicles' "Book XIV" of Euclid, whose icosahedron pyramids are the Cleo-Icos [7, 19]; the Cleo-Icos is also the unit tetrahedron of Mackay's icosahedron [7, 12]. The Fibonacci numbers and the golden ratio are classical.
- **Already published by DICTO.** The UP rule, its four stations and its first shells' pieces, and the chain's self-polarity [1]; the OG [3] and its downward family [2]; the OTT [5]; CLEO [6]; the CLEO sphere and the Cleo-Icos [7] (all listed in [8]). Shells 1 and 2 of the Cleo Towers and shell 4 of the Polar Cleo Towers are shells of [1].
- **Claimed here:** Theorems 1 and 2 (the containment bound and the icosahedral constant $15 - 6\sqrt{5} < \varphi$, with the covering lemma); Corollary 1 (the OG turned out is φ^k times the

downward family and nested for every k); Lemma 3 and Corollary 2 (polarity exchanges the reverse step and the UP step; the polar sequence of the Hecatodihedron); Theorem 3 read beat by beat with the Fibonacci boundaries; Lemmas 1 and 2 as used here; Proposition 1 (the ribbons cause the first break of the downward family, whose 60 outlying corners were found in [2]); Propositions 4 and 5, Conjecture 1 and Observation 1; Theorem 4 and Corollary 7 (Version 2); the exact towers of the fused shells (Cleo Towers, shells 3–5), of the Polar Cleo Towers, of the OG turned out (shells 1–3) and of its polar; the OG core of 80 Aktis and the OTT core of 38 OTT rays, with their shares, and the Oktaktis as Cleo-Icos lying in the OG's hexagon planes; the surface patterns and edge networks of the boundary solids (Table 5), including the TQH's 30 ribbons. Known and used: the (2, 3, 5) Möbius triangle and its side lengths, and polar reciprocation of polytopes [17].

In the sources checked (the full text of the 77 library sources with a text layer listed in the appendix, searched on 21 October 2026; the library's link-only entries were not searched) we did not find Theorems 1 or 2, Lemma 3, tower shells in Fibonacci beats, the Polar Cleo Towers, the Aktis or ray cores of the OG and the OTT, the OG turned out, or the ribbons of the TQH.

11. Further work

- Shell 5 of the OG turned out (5 beats, to beat 12), exact (§7): its towers (the nesting is Corollary 1).
- The full flip (tip = 2 × foot point) built exactly, and its shells.
- The surface patterns and lines of the shells 4 and 5 boundaries of the OG turned out, when built.

Outlook (DICTO's idea, untested). DICTO notes that these shells may mirror material structures, as geodesic forms did before C_{60} [24] and carbon nanotubes [25] were found. Nested icosahedral shells do occur in nature: in the clusters of quasicrystals and their approximants (Mackay [12], Bergman [22], Tsai [13]), in metal and boron nanoclusters, and in virus capsids (Caspar and Klug [23]; see also [26]). Whether the shells of this paper correspond to any real structure has not been tested; a checkable next step is to compare their corner counts and radius ratios with published sequences of cluster shells.

Data and reproducibility

Exact ($\mathbb{Q}(\sqrt{5})$), with the tools of [1] (exact_cycle.py, exact_shells.py, exact_downward.py in studies/2026-10-19-five-fold-octet) and of [4] (q5.py, exactpoly.py in studies/2026-10-18-ocd-octet/scripts): tower_shells_exact.py (the Cleo Towers; log tower-shells-exact.log, 6 PASS; data tower-shells-exact.json) and polar_exact.py (the polarity and the Polar Cleo Towers; polar-exact.log, 8 PASS; polar-exact.json) in 2026-10-20-cleo-towers; og_spires_exact.py (the 80 Aktis; og-spires-exact.log, 7 PASS) in 2026-10-20-og-spires; ott_rays_exact.py (the 38 rays; ott-rays-exact.log, 5 PASS) and ott_og_planes.py (the octahedral cuts in the OG's hexagon planes; ott-og-planes.log, 4 PASS, also the hexagon touching its triangle), on the OTT of [5] (cut8-exact.json, cut8-exact.log), in 2026-10-20-ott-rays; golden_exact.py (the OG turned out; golden-exact.log, 15 PASS and one FAIL line, which is only the test "one shape per kind of face" for shell 2, where the triangle towers have two shapes) and golden_orbits.py (shell 2 by symmetry orbit: two triangle orbits of 60 and one pentagon orbit of 12, one volume each; golden-orbits.log, 6 PASS) golden_ribbons.py (the trapezoids of the TQH, TQP and TQHD and

how they join; golden-ribbons.log, 14 PASS), surface_patterns.py (faces, kinds and groups of the nine boundary solids; surface-patterns.log, 28 PASS) surface_lines.py (their edge networks; surface-lines.log, 12 PASS) and kyushindo_checks.py (the condition w·v ≤ w·w for every station, and the 60-rotation symmetry of the inner and outer solid of every shell; kyushindo-checks.log, 13 PASS), in 2026-10-21-og-turned-out. Also in 2026-10-21-og-turned-out: nest_feet.py (foot points against their faces, and the nesting of the turned-out and downward sequences, k ≤ 8; nest-feet.log, 17 PASS), ribbons_return.py and axis_hull.py (axis words on the hulls and the faces at the two-fold edges, OG to down 9; ribbons-return.log, 14 PASS, axis-hull.log), polar_turned.py (Corollaries 2 and 6; polar-turned.log, 13 PASS), with polar_figs.py and polar_compose.py; axis_words.py (by the study's referee; its corner test runs over all listed foot points, so from down 6 on its results are used here only where they do not depend on that, as for down 10's faces; axis-words.log, axis-words-10.log). The proofs of Theorems 1 and 2 and the exact margins: studies/2026-10-21-nesting-proof (README.md; tmin_exact.py, tmin-exact.log and tmin-exact.json, k ≤ 8; covering_lemma.py and covering-lemma.log; tmin_general.py). The downward solids beyond down 4: golden_exact_long.py (golden-long-solids.pkl, on the node; down 12 still building). Float: flip_float.py (flip-float.json). Each script takes the path of its tool folder as its argument (python3 <script> <tool folder>); polar_exact.py chooses the corner triples by an exact determinant where the published helper of [1] used a float test. Figures: *_figs.py, shots.mjs and *_compose.py in each folder (Figures 8–10: surface_figs.py and surface_compose.py, which also draws the line figure) (float copies of the exact solids); the cover: cover.py in this paper's folder.

The series name

Kyushindo is the martial art of Kenshiro Abbe [20, 21]. DICTO gives its three principles as Change 万物流転, Rhythm 律動 and Harmony 調和, and his brief for this series was "rhythm, harmony and change in the construction of towers and cores". Here rhythm is the Fibonacci beats, harmony the exact golden fits, and change that each shell's form differs from the last (§7).

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4. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016 (the frame, and the exact tools q5.py and exactpoly.py).
5. DICTO, The DICTO Ogdoecontahedron on the Gyrated Octet, DICTO DOI: 10.5281/zenodo.23484012 (the DICTO OTT).
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27. P. Delsarte, J. M. Goethals, J. J. Seidel, Spherical codes and designs, Geom. Dedicata 6 (1977) 363–388, doi:10.1007/BF03187604.
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Appendix A. Computed tables

The exact values behind the constructions of §§3–8; every number is checked against the logs by audit.py.

Shell	Beats	From → to	Towers	Each tower (exact; ≈)	Shell volume
1	1	ID → RT	20 on triangles 12 on pentagons	(45 − 17√5)/120 ≈ 0.058224 5/12 ≈ 0.416667	(75 − 17√5)/6 ≈ 6.1645
2	1	RT → GTI	30 on rhombi	(3 − √5)/4 ≈ 0.190983	(45 − 15√5)/2 ≈ 5.7295
3	2	GTI → c·ID	20 on hexagons 12 on pentagons	(−236973 + 105982√5)/12 ≈ 0.829699 (1164825 − 520925√5)/24 ≈ 0.053701	(1124745 − 502955√5)/6 ≈ 17.2384
4	3	c·ID → c·FPD	20 on triangles 12 on pentagons	≈ 0.998686 ≈ 3.613281	≈ 63.3331
5	5	c·FPD → c³·ID	60 on triangles	≈ 5.135315	≈ 308.1189

Table 1. The Cleo Towers, at ID circumradius φ. Shell 3 passes through the FPD, shell 4 through c·RT and c·GTI, shell 5 through c²·ID, c²·RT, c²·GTI and c²·FPD. The exact values of shells 4 and 5 (denominators 261, 9 and 27) are in tower-shells-exact.json.

Shell	Beats	From → to	Towers	Each tower (exact; ≈)	Shell volume
1	5	RT/c² → GTI/c	30 on rhombi	(155284 + 69441√5)/1500000 ≈ 0.207039	(155284 + 69441√5)/50000 ≈ 6.2112
2	3	GTI/c → RT	12 on pentagons 20 on hexagons	(51911 − 17827√5)/83520 ≈ 0.144260 (146143 + 12334√5)/348000 ≈ 0.499203	(6343 − 741√5)/400 ≈ 11.7152
3	2	RT → FPD	30 on rhombi	(12982 − 5800√5)/27 ≈ 0.474286	(129820 − 58000√5)/9 ≈ 14.2286
4	1	FPD → c·ID	60 on triangles	(77875 − 34825√5)/27 ≈ 0.145655	(1557500 − 696500√5)/9 ≈ 8.7393
5	1	c·ID → c·RT	20 on triangles 12 on pentagons	(334125 − 149425√5)/3 ≈ 0.180821 (201250 − 90000√5)/3 ≈ 1.294008	(9097500 − 4068500√5)/3 ≈ 19.1445

Table 2. The Polar Cleo Towers, in the same frame; every kind of tower one shape, every shell filled exactly.

Rays	Base	Each ray (exact; ≈)	Height ≈	Share of the OTT
8 Oktaktis	equilateral triangle (octahedron cut)	$(-4 + 4\sqrt{5})/81 \approx 0.061040$	0.6228	36.78%
24 Triaktis	half-rhombus	$(-16 + 8\sqrt{5})/81 \approx 0.023315$	0.6667	42.15%
6 Tetraktis	whole rhombus	$(-32 + 16\sqrt{5})/81 \approx 0.046631$	0.6667	21.07%

Table 3. The OTT core, in the frame of [4, 5]. The Triaktis and Tetraktis have one height because all 30 of their bases lie in the triacontahedron's face planes.

Beat	Solid	Corners	Corner radii ≈
0	OG	102	1.618034, 1.687285
1	φ·TPH	80	2.445695, 2.564186
2	φ^2·TQH	132	3.896825, 3.933014
3	φ^3·TQP	140	6.003187, 6.032557, 6.324828
4	φ^4·TQHD	162	9.248108, 9.555175, 9.560121

Table 4a. The beats of the OG turned out built so far with towers (frame of [2]): corners and corner radii.

Shell	Beats	From → to	Towers	Each tower ≈	Shell volume ≈
1	1	OG → φ·TPH	60 on kites 20 on hexagons	0.349923 1.282073	46.6369
2	1	φ·TPH → φ^2·TQH	60 on triangles 60 on triangles 12 on pentagons	1.003994 0.992512 3.727717	164.5229
3	2	φ^2·TQH → φ^4·TQHD	60 on triangles 60 on trapezoids 20 on hexagons	36.321214 13.483680 6.977223	3127.8381

Table 4b. The OG turned out, golden step, in the frame of [2]; shell 3 passes through φ^3·TQP. All values are exact in golden-exact.json (for instance a hexagon tower of shell 1 is $(143097 + 71747\sqrt{5})/236748$); each shell's towers add up to the shell with remainder exactly 0.

Solid	Faces, and how faces of one kind group (exact)	Lines: edges; directions (exact)
ID	20 equilateral triangles, apart, on the three-fold axes; 12 regular pentagons, apart, on the five-fold axes	60; 15 directions of 4 edges, all along two-fold axes; 6 central belts of 10 edges
RT	30 rhombi, all joined	60; 6 directions of 10 edges, along the five-fold axes
GTI	12 regular pentagons, apart, on the five-fold axes; 20 hexagons, all joined	90; 15 directions of 6, along two-fold axes
FPD	60 isosceles triangles, all joined	90; 45 directions of 2 (15 along two-fold axes)
OG	60 kites in 12 groups of 5 on the five-fold axes; 20 regular hexagons, apart, on the three-fold axes, each bordered by kites on all six edges	180; 60 directions (30 of 2, 30 of 4), none along an axis
TPH	60 isosceles triangles, apart; 60 isosceles triangles in 30 pairs on the two-fold axes; 12 regular pentagons, apart, on the five-fold axes	210; 75 directions (15 of 6 along two-fold axes, 60 of 2)
TQH	60 isosceles triangles in 12 groups of 5 on the five-fold axes; 60 isosceles trapezoids in **30 ribbons** on the two-fold axes; 20 regular hexagons, apart, on the three-fold axes, each bordered by trapezoids on all six edges	270; 75 directions (15 of 6 along two-fold axes: the ribbons' lines; 30 of 2, 30 of 4)
TQP	60 isosceles triangles, apart; 30 rhombi, apart, on the two-fold axes; 60 isosceles trapezoids in 12 rings of 5 on the five-fold axes; 12 regular pentagons, apart, on the five-fold axes	300; 75 directions (15 of 8 along two-fold axes, 30 of 2, 30 of 4)
TQHD	60 isosceles triangles, apart; 60 isosceles triangles in 12 groups of 5 on the five-fold axes; 20 regular hexagons, apart, on the three-fold axes; 60 isosceles trapezoids in 12 rings of 5 on the five-fold axes	360; 105 directions (15 of 8 along two-fold axes, 60 of 2, 30 of 4)

Table 5. The boundary solids. "Apart": no two faces of that kind share an edge. A group is "on" an axis when a rotation about that axis maps it to itself. A direction is "along" an axis when its edges are parallel to it. A central belt is a closed circuit of edges in one plane through the centre.

Axis	OG	TPH	TQH	TQP	TQ HD	down 5 TQP-252	down 6 TQ-320	down 7 TQP-402	down 8 TQ-500	down 9 TQD-732
five-fold	corner	face	corner	face	corner	face	corner	face	corner	face
three-fold	face	corner	face	corner	face	corner	face	corner	face	corner
two-fold	corner	**edge**	**edge**	face	corner	**edge**	**edge**	face	**edge**	**edge**
at the two-fold edge		triangle pair	**ribbon**			triangle pair	trapezoid pair (longer sides)		**ribbon**	**ribbon**

Table 6. What each axis leaves the solid through, for every solid of the OG's downward family built (exact). At a two-fold edge: "triangle pair", two congruent isosceles triangles meet there; "ribbon", two congruent isosceles trapezoids meet across their shorter parallel sides and all 60 trapezoids of that kind pair so, as on the TQH; at down 6 two congruent isosceles trapezoids meet across their longer parallel sides.

k	Corners (listed)	Faces	$t_{min}(D_k)$ exact	≈	$\varphi - t_{min}$
0	102	80	$(445 - 16\sqrt{5})/361$	1.133581	+0.4845
1	80	132	$(5941 - 1622\sqrt{5})/2071$	1.117382	+0.5007
2	132	140	$(247111 - 104650\sqrt{5})/11881$	1.103147	+0.5149
3	140	162	$(9 - 3\sqrt{5})/2$	1.145898	+0.4721
4	162	200	$(21 - \sqrt{5})/18$	1.042441	+0.5756
5	200	252	(long, in tmin-exact.json)	1.086902	+0.5311
6	192 (252)	320	(long, in tmin-exact.json)	1.025657	+0.5924
7	260 (320)	402	(long, in tmin-exact.json)	1.059516	+0.5585
8	312 (402)	500	(long, in tmin-exact.json)	1.138077	+0.4800

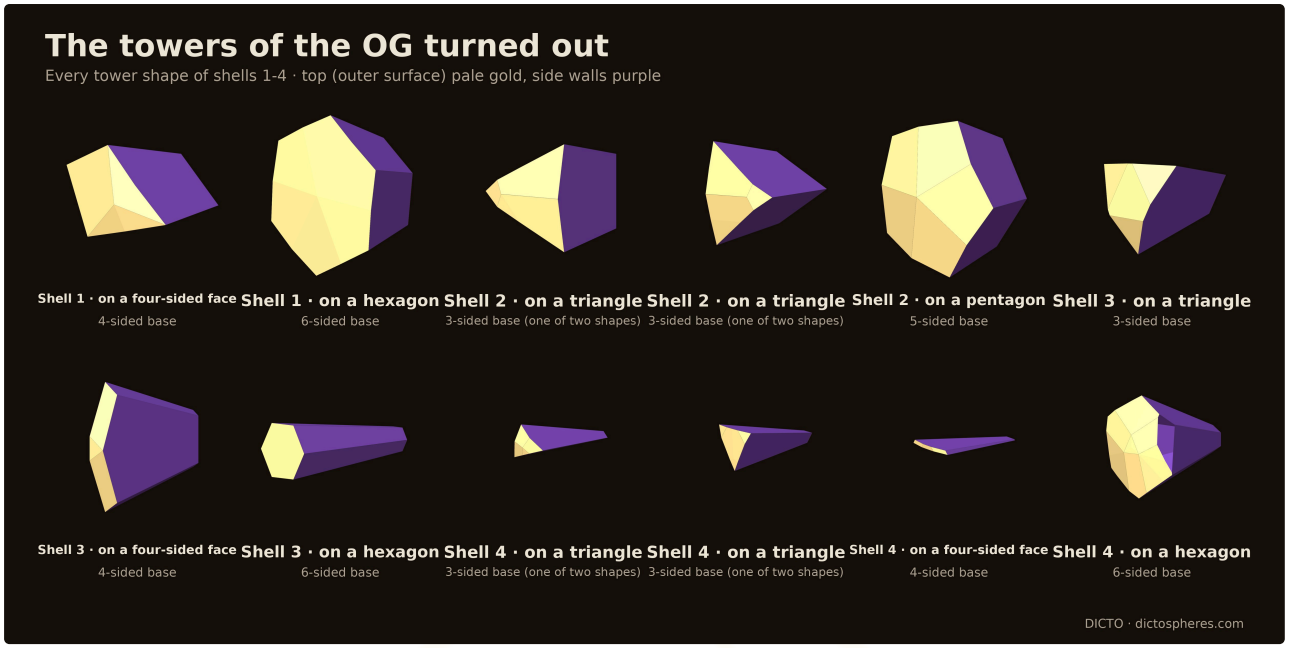
Table 7. The containment margins of the OG's downward family (tmin_exact.py, frame of [2]): every value exact in $\mathbb{Q}(\sqrt{5})$ and below φ by exact sign; the largest is $t_{min}(D_3) = (9 - 3\sqrt{5})/2 = 3/\varphi^2$. Not computed for k ≥ 9.

Shell	Beats	Inner inside outer	Towers (on the inner solid)	Each tower ≈	Shell volume ≈
1	1	polar S_1 inside polar S_0	60 on 5-gons 20 on 6-gons	3.760911 0.691479	239.4842
2	1	polar S_2 inside polar S_1	60 on 3-gons 60 on 5-gons 12 on 5-gons	0.370515 0.427719 1.121247	61.3490
3	2	polar S_4 inside polar S_2	60 on 4-gons 30 on 4-gons 12 on 5-gons 60 on 5-gons	0.083893 0.259256 0.000008 0.152786	21.9785

Table 8. The polar of the OG turned out (Corollary 6), at φ^4 times the frame of reciprocation about the unit sphere in the frame of [2]; one shape per orbit of faces, each shell's towers adding up to it with remainder exactly 0 (exact values in polar-turned.json).

Solid	Name (DICTO)	Corners (listed points)	Faces	Face kinds
down 5	DICTO TQP-252	200 (200)	252	120 triangles, 120 quadrilaterals, 12 regular pentagons
down 6	DICTO TQ-320	192 (252)	320	240 triangles, 20 regular triangles, 60 quadrilaterals
down 7	DICTO TQP-402	260 (320)	402	300 triangles, 90 quadrilaterals, 12 regular pentagons
down 8	DICTO TQ-500	312 (402)	500	360 triangles, 20 regular triangles, 120 quadrilaterals
down 9	DICTO TQD-732	500 (500)	732	540 triangles, 180 quadrilaterals, 12 decagons
down 10	DICTO TQHD-662	480 (732)	662	480 triangles, 150 quadrilaterals, 20 hexagons, 12 decagons
down 11	DICTO TQP-672	470 (662)	672	420 triangles, 240 quadrilaterals, 12 regular pentagons

Table 9. The new solids of the OG's downward family and DICTO's names (face kinds and face count). Points exact; hull faces and face kinds measured on the exact points by float tests (new_solids_faces.py), not yet proved exactly. The TQHD-662 is not the TQHD of [2], which has 200 faces.



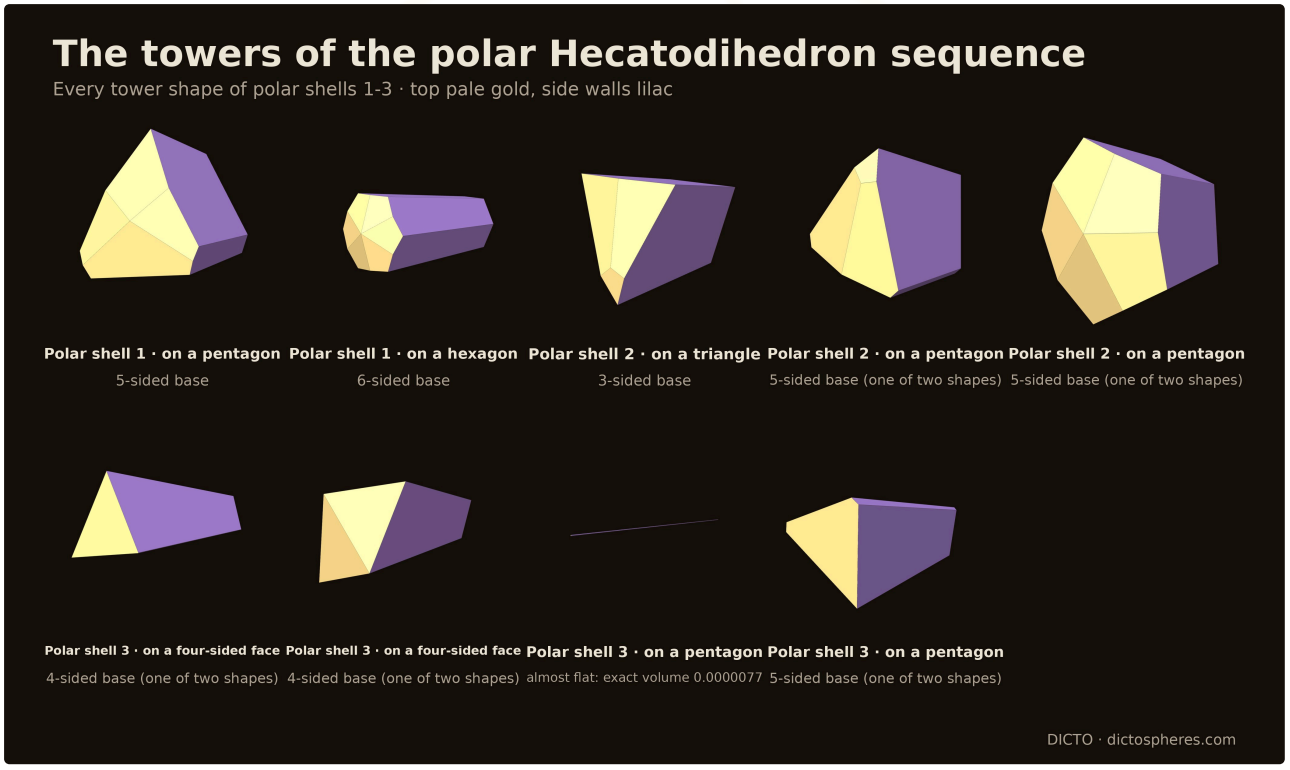


Figure 11. The tower catalogue (DICTO: the shells are new configurations too). Every tower shape, one of each kind: above, the OG turned out, shells 1 to 4 (2, 3, 3 and 4 shapes; 12 in all); below, the polar Hecatodihedron sequence, shells 1 to 3 (2, 3 and 4 shapes; 9 in all). Each tower is the cone from the centre over one face of the inner solid, cut by the outer solid; top (on the outer solid) pale gold, side walls purple (lilac for the polar towers). The 12 pentagon towers of polar shell 3 are almost flat (exact volume 7.65×10^{-6}). Shapes grouped by exact volume as in the shell logs; float renders of the exact boundaries.

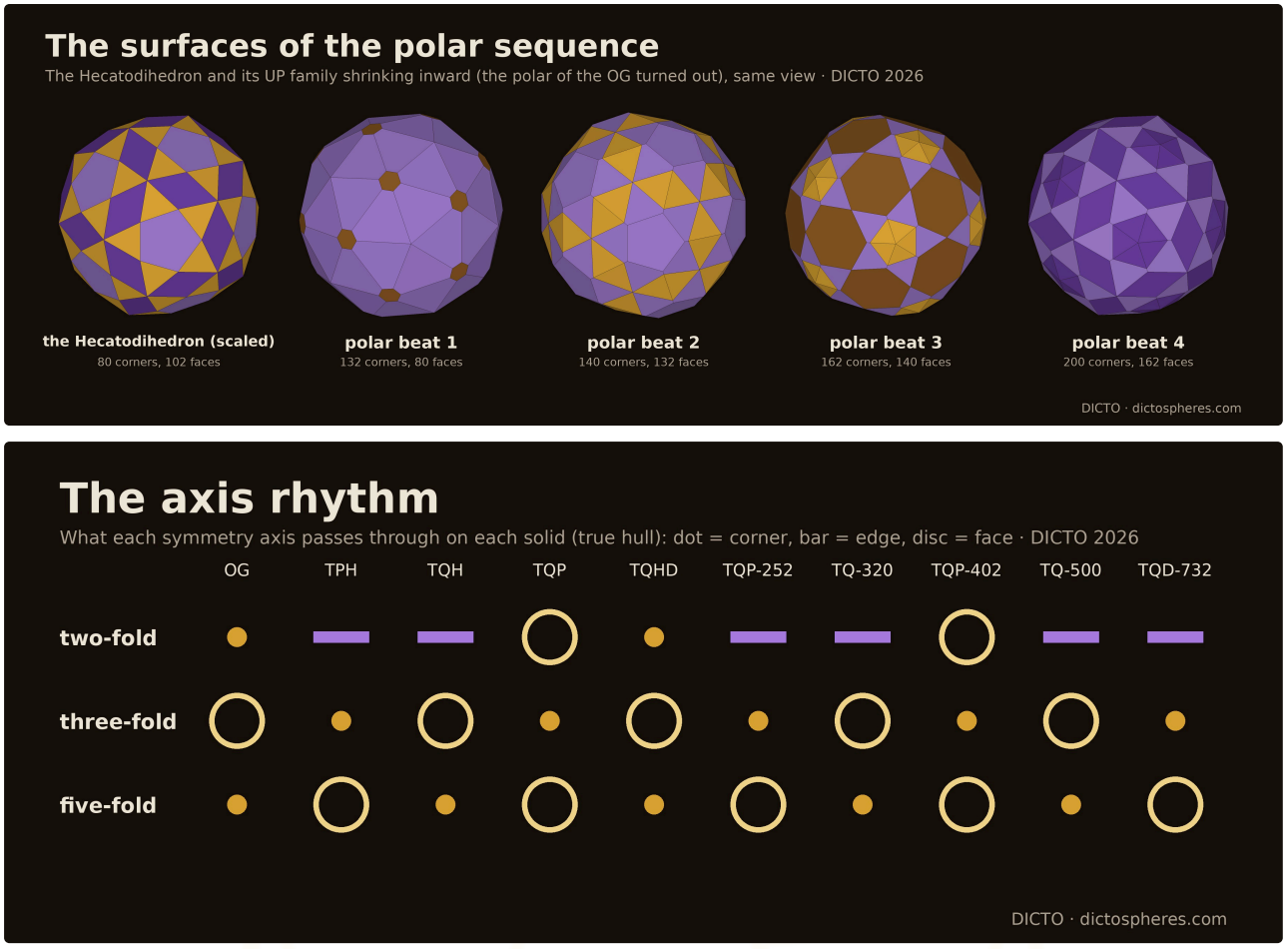


Figure 12. Above: the surfaces of the polar sequence (Lemma 3, Corollary 2): the Hecatodihedron (scaled) and polar beats 1 to 4, faces coloured by kind (triangles gold, four-sided purple, pentagons lilac, hexagons bronze), same view as Figure 9. Below: the axis rhythm (Proposition 5, Observation 1): what each two-, three- and five-fold axis passes through on the true hull of each solid, from the OG to DICTO TQD-732 (beat 9): dot a corner, bar an edge, disc a face. The three- and five-fold rows alternate face and corner at every beat; the two-fold row repeats corner, edge, edge, face from the OG to beat 7 and breaks at beat 8. Float drawings of the exact points.

Table 10. $t_{min}(D_k)$ for $9 \le k \le 40$ (Theorem 5): exact in $\mathbb{Q}(\sqrt{5})$ for $k \le 13$, certified upper bounds (shown to 6 places) beyond; all $< 3/\varphi^2 \approx 1.145898$. The last column is the number of faces of D_{k+1}.

k	$t_{min}(D_k)$	how	faces of D_{k+1}
9	1.076770	exact	–
10	1.008772	exact	672
11	1.018734	exact	902
12	1.020352	exact	1092
13	1.027602	exact	1262
14	1.024833	certified	1392
15	1.021610	certified	2360
16	1.011525	certified	2882
17	1.011068	certified	3780
18	1.008139	certified	5612
19	1.006896	certified	7952
20	1.005365	certified	11322
21	1.003401	certified	15500
22	1.003051	certified	25262
23	1.002128	certified	33980
24	1.002565	certified	47052
25	1.001215	certified	66482
26	1.000652	certified	89742
27	1.000671	certified	110672
28	1.000769	certified	152112
29	1.000677	certified	199382
30	1.001014	certified	264642
31	1.000273	certified	345170
32	1.000260	certified	424142
33	1.000187	certified	532262
34	1.000146	certified	739862
35	1.000113	certified	1034522
36	1.000143	certified	1411742
37	1.000083	certified	1873742
38	1.000098	certified	2533472
39	1.000083	certified	3412382
40	1.000108	certified	4517522

Appendix B. The reference library searched

The library's manifests list 261 entries, of which 90 have a local file; with three files at its root, these give the 77 distinct sources with a text layer listed below and 13 page images. The full text of the 77 sources was searched on 21 October 2026 for: pedal, antipedal, reciprocation, polar, nested, concentric, shell, Fibonacci, central pyramid, spire, tower, ogdoecontahedron, octahedrally truncated, Hypsicles, Mackay and Kyushindo. None contains the results of this paper. The relevant hits are Kepler's nested solids [11]; the nested icosahedral point sets of Dechant, Wardman, Keef and Twarock and of Twarock, Valiunas and Zappa [14, 15]; the concentric cluster shells of Fujita, Takano, Yamamoto and Tsai [13];

Fibonacci sequences in Baake's guide and the other aperiodic-order surveys; and the Wikipedia pages on Kyūshindō and Kenshiro Abbe [20, 21]. The 13 page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885) and the about 170 link-only entries were not searched.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov (chapter 13, English translation)	1885	Elements of the Theory of Figures, chapter 13
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
R. V. Moody	2000	Model sets: a survey
I. Hafner, T. Zitko	2002	Rhombic polyhedra: from dissection of the cube to space filling with prolate rhombohedra and rhombic dodecahedra of the second kind
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2007-2015	Wolfram Demonstrations: 54-Faced Space-Filling Polyhedron (2007), Golden Octet Truss (2007), A Rhombic Dodecahedron Made of Rhombic Triacontahedra (2008), Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (2008), Cluster of 14 Rhombic Triacontahedra (2015)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)

J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures

N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)